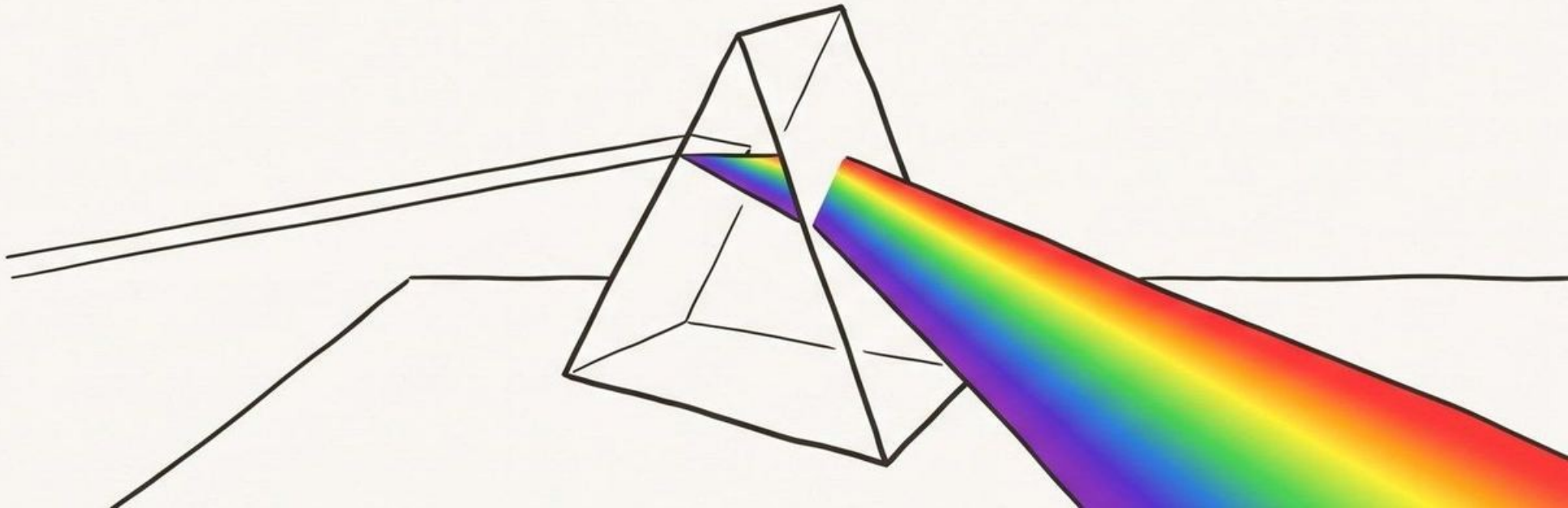
HANDS-ON WORKSHOP

Two exercises · bring a screen

How Color Works

A 90-minute hands-on workshop

One beam goes in. **Everything you argue about comes out.**



THE CONTRACT

By the end of this session you will be able to

1. Name why a specific color pair **fails to read**
2. Check any pair against a **WCAG threshold** and read the result out loud
3. Pull a **five-color palette** off a real image, then repair the pair that fails
4. Say why a **rainbow ramp** is the wrong default for ordered data

This session simplifies in two places, and says so where it does.

The 90 minutes

0–10	Why color is not taste
10–35	The minimum model
35–50	Worked demonstration
50–62	**Your turn (12 min)**
62–90	Mistakes, check and close

Bring a screen you can pick colors on.

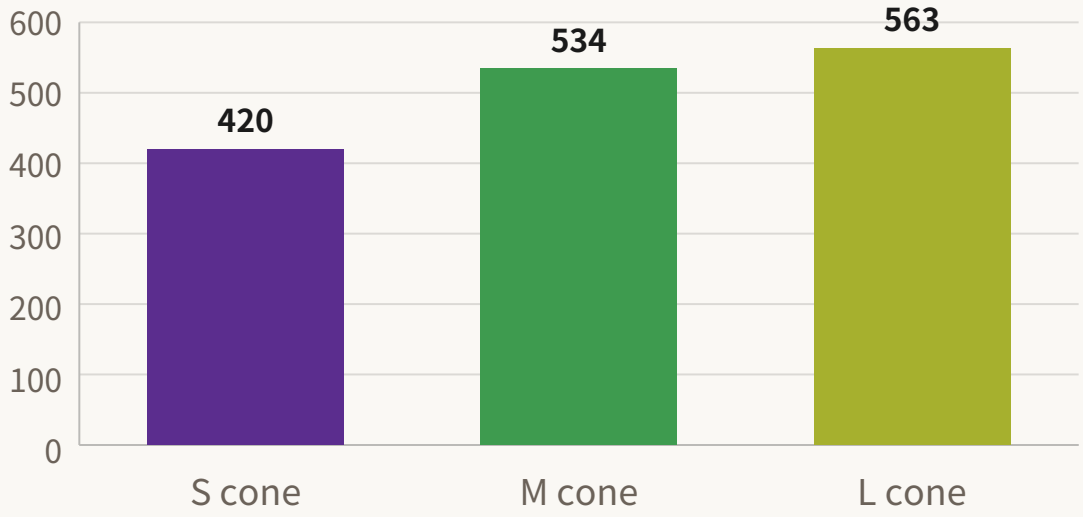
Your eye does not measure light

It samples the spectrum three times.
Everything you will ever see is rebuilt from **those three numbers**.

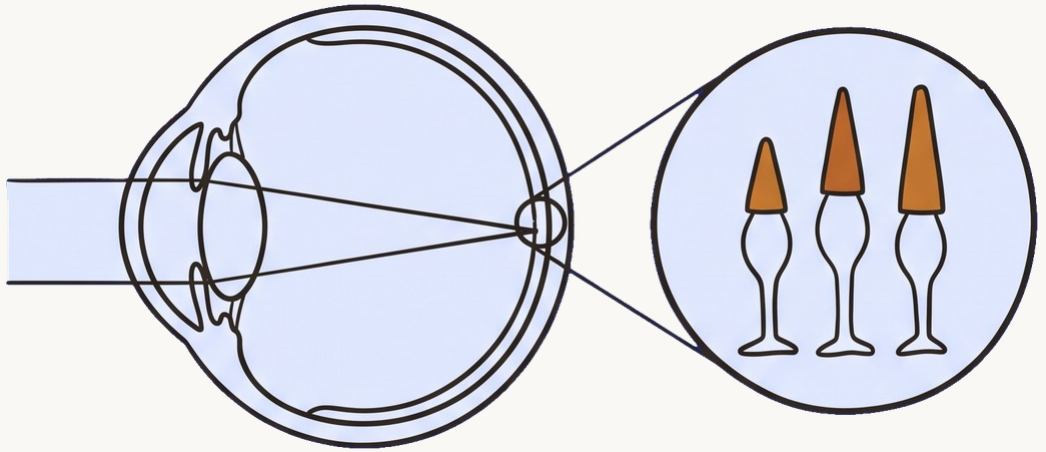


≈380 nm ≈700 nm

CONVENTION Sensitivity fades out at both ends, and different authorities round the endpoints differently.



RULE Human vision is trichromatic: S, M and L.



Peaks: Bowmaker & Dartnall 1980 · retina counts: NCBI Bookshelf · full curves: Stockman & Sharpe via cvrl.org

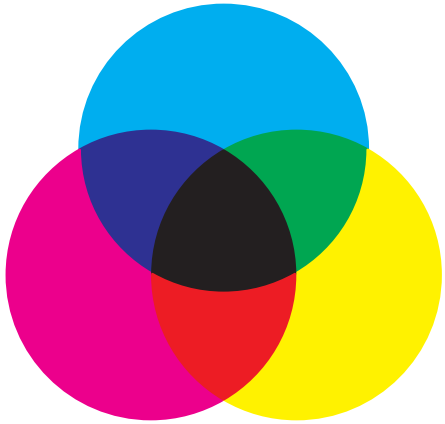
Two ways color mixes — and you ship into both

ADDITIVE · LIGHT · EVERY SCREEN

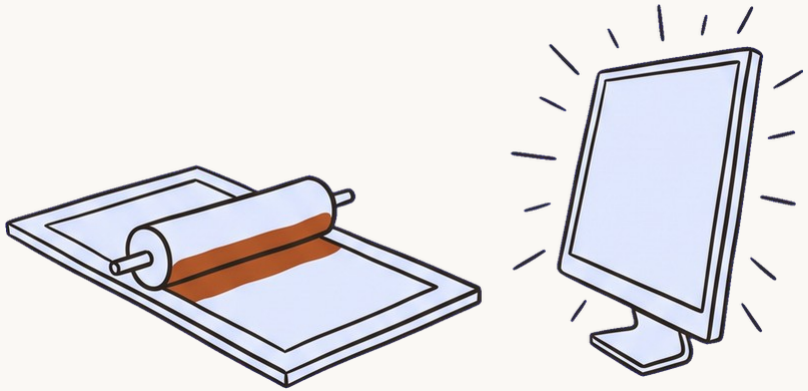


Start at black. Adding light moves toward white.

SUBTRACTIVE · INK · ALL PRINT



Start at white. Adding ink moves toward black.



RULE Two machines, two directions.

The two primary sets are **complements** of each other: each additive primary is a secondary in the other system, and the other way round.

A screen color need not be reachable in ink.

Additive / subtractive systems: RMIT Colour Theory 1 (open textbook)

MINIMUM CONCEPT · 3 OF 3

Three dials — schemes turn only one of them

HUE — an angle, 0–360°

red 0 / 360 · green 120 · blue 240

SATURATION — 0% gray to 100%

how much of the hue is left in the color

LIGHTNESS — 0% black to 100% white

the only dial the contrast rule looks at

CONVENTION harmony schemes

The scheme names come from Johannes Itten’s 1961 Art of Color, taught at the Bauhaus. Munsell drew a different wheel — so which colors count as “complementary” depends on which wheel you started from.

Itten’s twelve-hue wheel: a teaching device, not a law of light.



complementary = two wedges 180° apart

HSL definition: W3C CSS Color 4 · Itten: Getty Bauhaus teaching materials · competing wheels: Munsell

Lightness is the dial that carries legibility

RULE WCAG SC 1.4.3

The W3C's text-contrast rule is defined purely by relative luminance. There is **no hue term** and no saturation term in it.

```
L = 0.2126 R + 0.7152 G + 0.0722 B
```

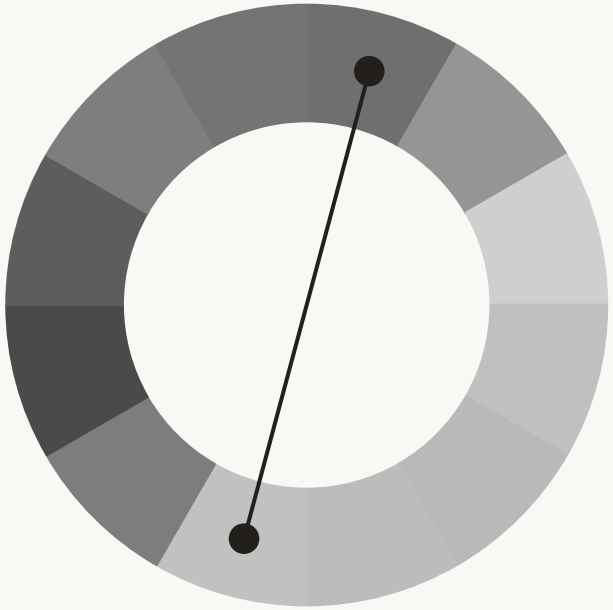
FAIL — far apart in hue, almost equal in luminance

Read this — 1.10:1

PASS — one hue, far apart in lightness

Read this — 9.90:1

The same wheel, with only the part the rule actually measures left.



the same two wedges:
2.78:1

Relative luminance and SC 1.4.3: W3C WCAG 2.1 · ratios computed from the swatches on this page

REFERENCE CARD

The numbers you actually have to hit

Five numbers. You can check any of them in about ten seconds.

TEXT OR ELEMENT CLASS	CRITERION	LEVEL AA	LEVEL AAA
Normal text	1.4.3 / 1.4.6	**4.5:1**	7:1
Large text	1.4.3 / 1.4.6	3:1	4.5:1
UI components and graphics	1.4.11	3:1	no higher level

RULE how the number is computed

Contrast ratio is (L1 + 0.05) / (L2 + 0.05) with the lighter luminance on top. It runs from 1:1 to 21:1, and must not be rounded before you compare it with a threshold.

What counts as large text

Large text means at least 18 pt, or at least 14 pt bold — the spec puts that at roughly 24 px and 18.5 px.

Five colors out of one print



Katsushika Hokusai, Under the Wave off Kanagawa, c. 1830–32. Public domain, via Wikimedia Commons.

Sampled in this order: the largest area first, then the darkest, then the lightest, then two accents chosen for meaning rather than area.

Sky paper	#E3DBCD	sky and paper ground
Aged ground	#C5B4A1	largest single area
Sea green	#6B8C8E	the wave body
Boat slate	#414D59	boats and Fuji shadow
Deep indigo	#060F2C	the trough shadows

Five samples. Not a palette — nothing measured.

Now measure it, then repair the pair that fails

Sky paper | Aged ground | Sea green | Boat slate | Deep indigo

Every pair, against the 4.5:1 body-text floor

	Sky paper	Aged ground	Sea green	Boat slate	Deep indigo
Sky paper	–	1.47	2.65	6.29	13.74
Aged ground	1.47	–	1.81	4.29	9.36
Sea green	2.65	1.81	–	2.37	5.18
Boat slate	6.29	4.29	2.37	–	2.18
Deep indigo	13.74	9.36	5.18	2.18	–

The one pair that fails, and its repair

Body text → Body text

4.29:1 FAIL

4.55:1 PASS

Same hue, same saturation. Only the lightness moved, from 30% down to 29%: `#414D59` becomes `#3E4955`. The palette still reads as the same palette.

PREFERENCE

Passing is not the same as good. Only usable.

Threshold: W3C WCAG 2.1 SC 1.4.3 · ratios computed from the five sampled values above

GUIDED PRACTICE

Your turn — 12 minutes

TASK

Open any image you own. Sample five colors in the order we just used, then measure the pair you would actually put together for body text on a background.

STARTING POINT

Your own image, and the contrast checker linked on the last page of this deck.

SUCCESS CONDITION

Five hex values written down, one measured ratio, and that ratio at `4.5:1` or above.

IF YOU GET STUCK

Stop sampling for meaning and sample by area instead: largest first, then darkest, then lightest.

12 minutes — then we compare three of them out loud.

COMMON MISTAKE · 1 OF 2

Mistake 1 — turning up the saturation

AS DRAWN	SATURATION PUSHED	LIGHTNESS LOWERED
Caption text	Caption text	Caption text
#C98E62	#FF862C	#A06437
2.63:1 FAIL	2.28:1 FAIL	4.53:1 PASS

The same three colors, reduced to what the rule actually measures:

L 0.33 L 0.38 L 0.17

PREFERENCE why it is tempting

On a big bright monitor the louder color really does look stronger to you.

RECOMMENDATION what to do instead

Move the lightness, keep the hue, then measure it again.

Ratios and relative luminances computed from the three swatches on this page, per W3C WCAG 2.1

COMMON MISTAKE · 2 OF 2

Mistake 2 — a rainbow ramp for ordered data

RAINBOW RAMP · FIVE ORDERED STEPS



0.02 0.17 0.26 0.73 0.16

relative luminance: up, up, up, then down

MONOTONIC RAMP · SAME FIVE STEPS



0.82 0.56 0.31 0.14 0.04

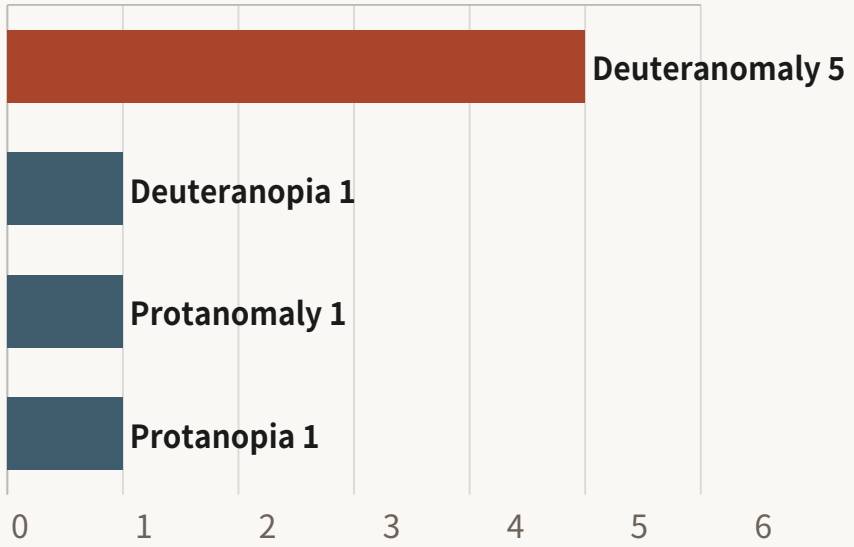
relative luminance: down, every step

RULE Borland & Taylor 2007: three documented defects

1. No perceptual ordering: equal-looking color steps are not equal data steps.
2. Uncontrolled luminance variation across hues obscures the data.
3. It can create gradients that are not in the underlying data at all.

RECOMMENDATION

Crameri et al. (2020): use a perceptually uniform map.



Rainbow maps: Borland & Taylor 2007; Crameri et al. 2020 · prevalence: NEI, Colour Blind Awareness

UNDERSTANDING CHECK

Check yourself — three questions

Q1

You have two colors measured at 3.8:1 and you need them for 16 px body text. What exactly do you change, and why that and not the hue?

Q2

A colleague says a palette is harmonious because it is triadic. Which part of that is a rule and which part is a convention?

Q3

You have a sequential heat map drawn in a rainbow ramp. Name two things that go wrong before anyone's vision is involved.

ANSWERS — ALL THREE ARE CHECKABLE AGAINST THIS DECK

4.5:1 is the AA floor at that size, so move the lightness until the measured ratio clears it. The formula has no hue term in it.

None of it is a rule. Triadic is an angular convention on the hue dial, and harmony is not a measurable property.

No perceptual ordering, and uncontrolled luminance variation across hues.

RECAP AND NEXT STEPS

What you can do now, and where to go next

Separate hue from lightness, measure the lightness, then argue.

YOU CAN NOW

- Name why a color pair fails to read
- Check any pair against a WCAG threshold
- Pull and repair a five-color palette
- Say why a rainbow ramp is the wrong default

WHERE TO GO NEXT

- — a ratio plus AA/AAA pass-fail
- — every criterion, filterable
- — any image under each type of deficiency
- — sequential, diverging, qualitative
- — themes and harmony generators
- — the cone-fundamentals data behind page 3

What this session simplified: the cone sensitivity curves overlap far more than three columns suggest, and a contrast ratio is a floor, not a measure of quality.

Every destination above is linked live in this file; sources for each claim are on the page that makes it.